

PREDICTIVE ANALYTICS
PROJECT REPORT
(Project Semester January-April 2025)

***ADVANCED VEHICLE ANALYTICS & TRAFFIC VIOLATION DETECTION
SYSTEM***

Submitted by

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Programme and Section – B.Tech CSE/IT, K23BE

Course Code – INT234

Under the Guidance of

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Discipline of CSE/IT

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Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Sushant Kumar bearing Registration no. 12311087 has completed INT234 project titled, “**Advanced Vehicle Analytics and Traffic Violation Detection System**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Dr. Tanim Thakur

Assistant Professor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 16 December, 2025

DECLARATION

I, Sushant Kumar, student of Bachelor of Technology under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 16 December, 2025

Sushant

Registration No. 12311087

Sushant Kumar

ACKNOWLEDGEMENT

I would like to express my deepest gratitude and sincere appreciation to my supervisor and mentor, Dr. Tanima Thakur, for her invaluable guidance and unwavering support throughout the development of the **"Advanced Vehicle Analytics Dashboard"**. Her profound insights into **Predictive Analytics** and **Machine Learning** were instrumental in shaping the direction of this research, particularly in refining the system's ability to generate strategic insights and actionable recommendations. Her constructive feedback on the implementation of advanced diagnostics and model robustness indicators significantly elevated the quality of this study.

I also extend my heartfelt thanks to **Lovely Professional University** for providing the essential infrastructure and state-of-the-art resources required to complete this comprehensive study. The access to high-performance computational facilities facilitated the rigorous training of our YOLOv8n models and the processing of complex video analytics data. The supportive academic environment provided by the university has been a constant source of motivation, enabling the successful realization of this project.

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INTRODUCTION

Overview of Predictive Analytics

Predictive analytics encompasses a variety of statistical techniques from data mining, predictive modeling, and machine learning that analyze current and historical facts to make predictions about future or otherwise unknown events. In the context of computer vision, predictive analytics is used to identify objects, forecast their trajectories, and detect anomalies in real-time environments.

Importance of Prediction in the Selected Domain

With the exponential rise in vehicular traffic, manual monitoring has become obsolete and error-prone. Predictive analytics in Intelligent Transportation Systems (ITS) is crucial for:

- **Accident Prevention:** Detecting overspeeding vehicles before accidents occur.
- **Traffic Flow Management:** Classifying vehicle types to optimize signal timings.
- **Automated Enforcement:** Reducing the burden on traffic police by automatically extracting number plates of violators.

Problem Statement

Current traffic monitoring systems are often passive, recording video without actionable insights. There is a lack of integrated systems that can simultaneously perform **Multi-Class Vehicle Detection, Speed Estimation, Anomaly Detection (Overspeeding)**, and **License Plate Recognition (OCR)** in real-time with high accuracy.

Objectives of the Study

1. To develop a predictive model capable of detecting and classifying vehicles into 7 distinct classes (Car, Truck, Bus, Auto, Two-Wheeler, Plate, Blur-Plate).
2. To implement an algorithm for real-time speed estimation and anomaly detection based on a configurable threshold (e.g., >80 km/h).
3. To integrate Optical Character Recognition (OCR) for reading license plates of violating vehicles.
4. To compare three variants of the YOLO model to identify the optimal balance between accuracy and inference speed.

Scope of the Project

The project is limited to daytime traffic conditions and utilizes computer vision techniques. It includes a frontend dashboard (Streamlit) for user interaction, a backend database (MySQL) for logging violations, and a reporting module for generating PDF analytics.

Expected Outcomes

A fully functional dashboard that takes video/image input and produces an annotated output with bounding boxes, speed labels, and a downloadable "Strategic Traffic & Safety Report."

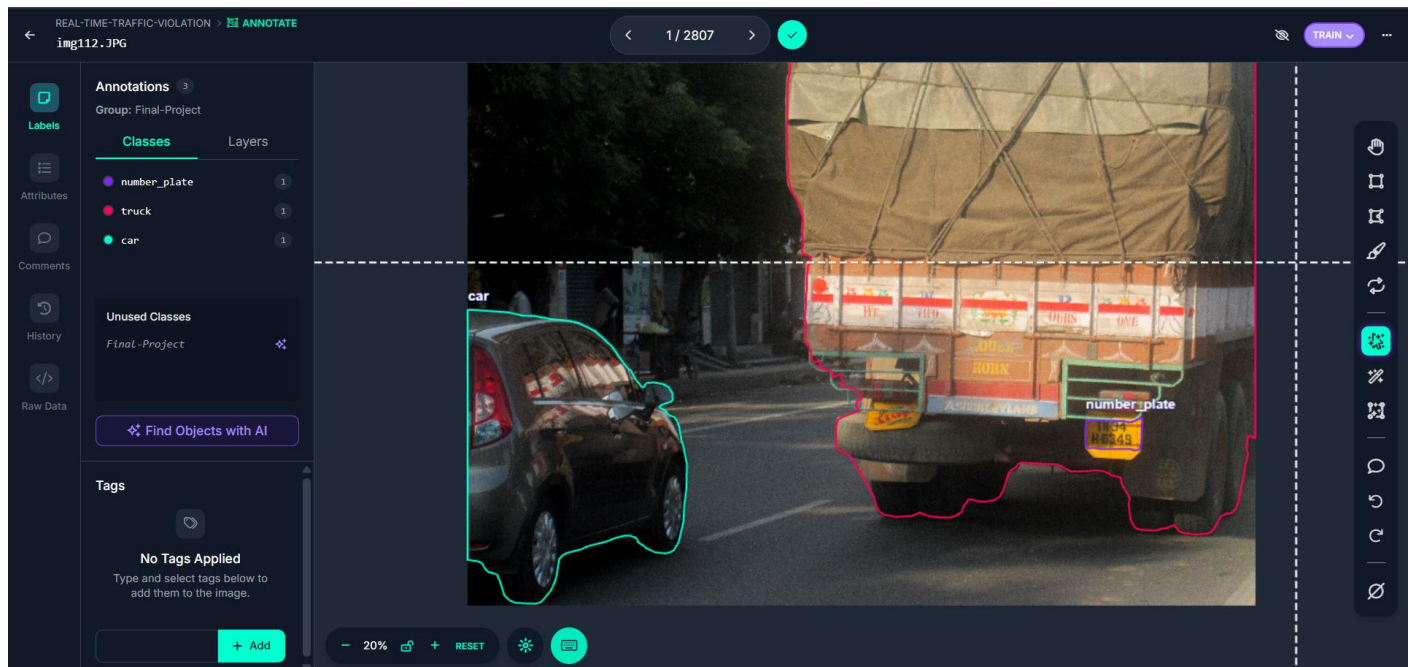
SOURCE OF DATASET

- **Name of Dataset:** Vehicle Detection & License Plate Dataset (v1).
- **Source Link:** Primary Source + Roboflow Universe + Kaggle
 - a. https://www.kaggle.com/datasets/saumyapatel/traffic-vehicles-object-detection?utm_source=chatgpt.com
 - b. <https://app.roboflow.com/code-jvba3>
- **Description of Attributes:**
- **Images:** .jpg files containing diverse traffic scenarios.
- **Labels:** .txt files (YOLO format) containing class IDs and normalized bounding box coordinates (x_center, y_center, width, height).
- **Classes:**
 - 0: Car
 - 1: Plate (License Plate)
 - 2: Blur_Plate
 - 3: Two Wheeler
 - 4: Auto
 - 5: Bus
 - 6: Truck
- **Number of Rows/Columns:** The dataset consists of approximately 960 images split into Training (772), Validation (127), and Test (61) sets.
- **Type of Prediction:** Multi-Class Object Detection (Regression for Bounding Boxes & Classification for Object Types).

DATASET PREPROCESSING

Before feeding data into the model, rigorous preprocessing was applied to ensure high model generalization:

1. **Resizing:** All input images were resized to **640x640 pixels** to match the YOLOv8 input layer requirements.
2. **Normalization:** Pixel values were normalized from $[0, 255]$ to $[0, 1]$ to accelerate gradient descent convergence.
3. **Data Augmentation:** To improve robustness, the following augmentations were applied during training:
 - **Mosaic Augmentation:** Stitching 4 images together to teach the model to detect small objects and handle context.
 - **HSV Scaling:** Random adjustments to Hue, Saturation, and Value to handle different lighting conditions (sunny, cloudy).
 - **Random Flip:** Horizontal flipping to simulate traffic moving in different directions.



Noise



0%



3.9%

Add noise to help your model be more resilient to camera artifacts.



Warning: These settings are outside of the typical range and may produce poor results.

Why would I add noise to my images? [🔗](#)

Noise can help defend against adversarial attacks and prevent overfitting.

via Roboflow Blog

Go Back

Apply

ANALYSIS ON DATASET

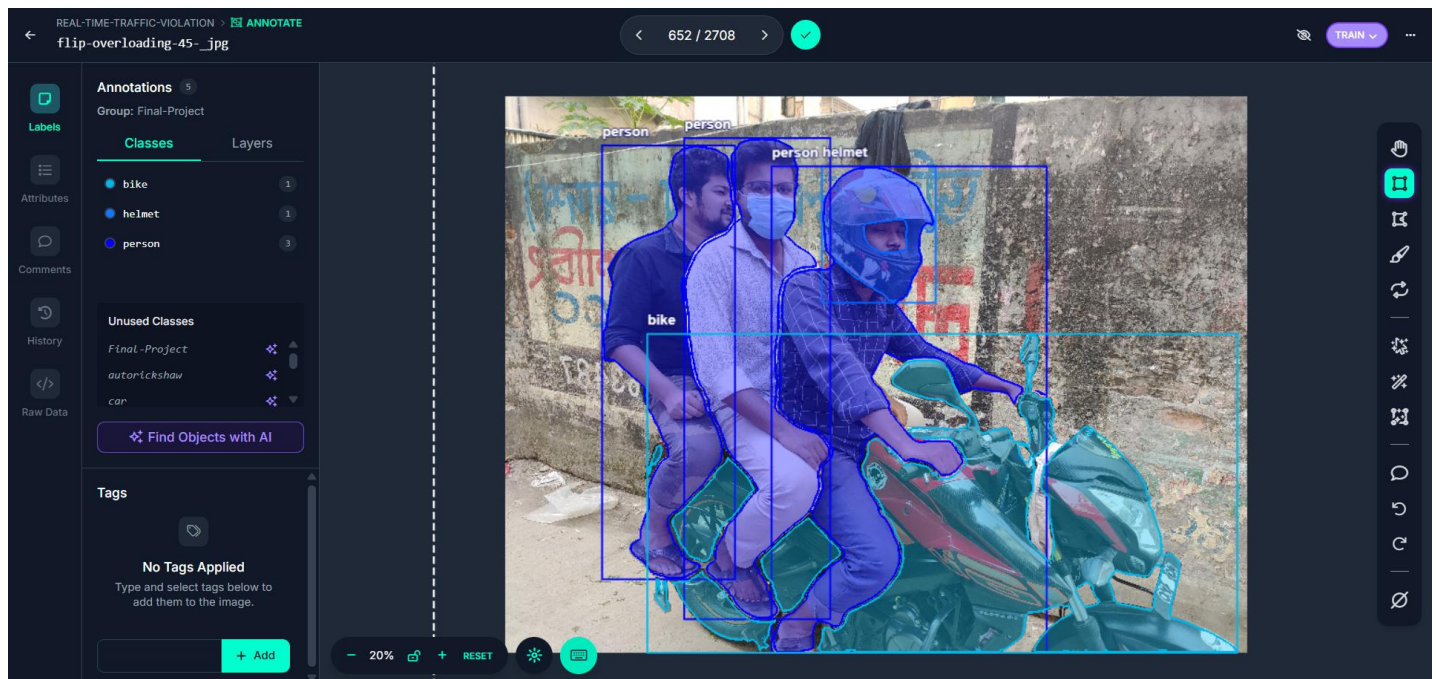
General Description

The dataset is heavily skewed towards "Cars" and "Two Wheelers," which mirrors real-world traffic composition. Special care was taken to include "Trucks" and "Buses" to ensure the model does not misclassify large vehicles.

Analysis Results & Visualization

We analyzed the distribution of bounding boxes to understand object scales.

- **Small Objects:** License plates (Classes 1 & 2) occupy less than 5% of the frame area.
- **Large Objects:** Buses and Trucks occupy significant frame area.



MODEL DEVELOPMENT

We utilized the **YOLOv8 (You Only Look Once)** architecture due to its state-of-the-art performance in real-time detection.

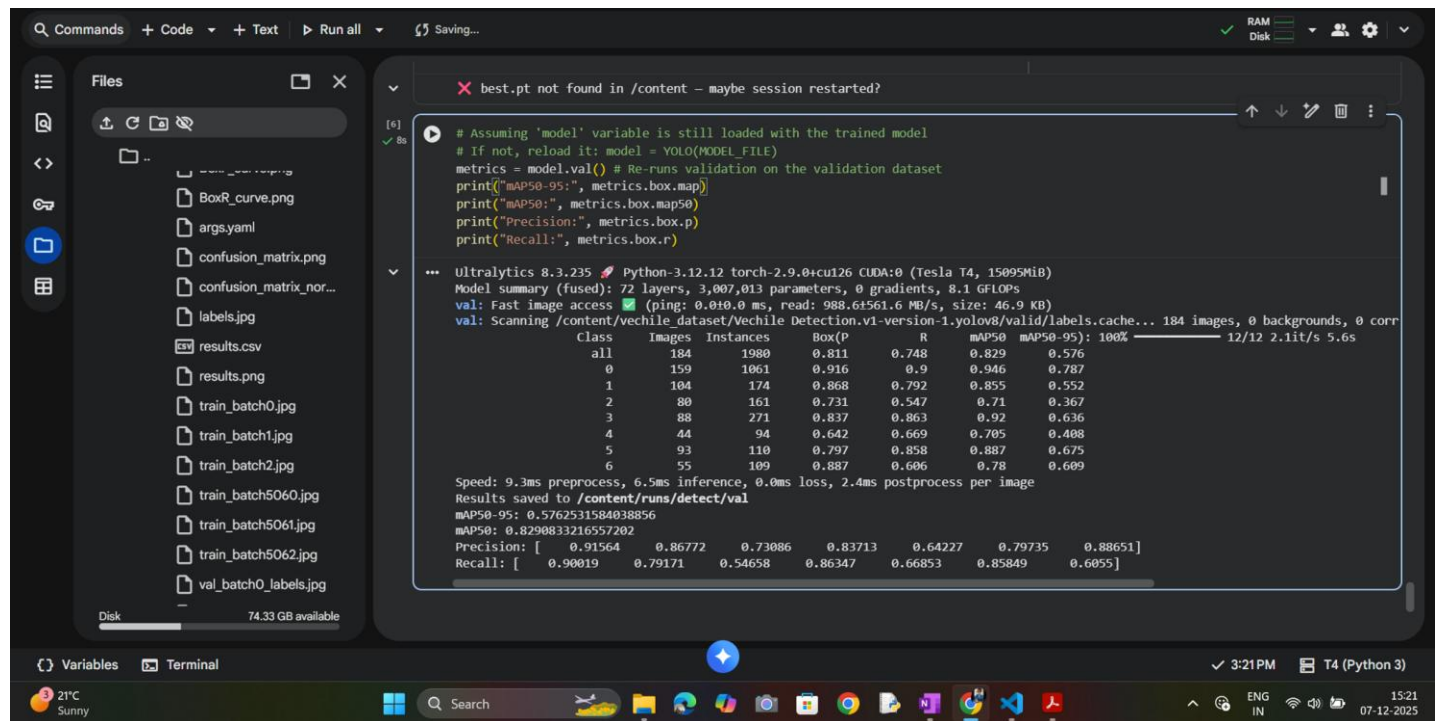
Training Configuration

To ensure maximum accuracy, we deviated from default settings by tuning the epochs.

- **Framework:** Ultralytics YOLOv8
- **Epochs:** Increased from default to **40 Epochs** to allow the model to converge fully without overfitting.
- **Batch Size:** 16
- **Image Size:** 640

Three specific models were trained to evaluate performance:

1. **YOLOv8n (Nano):** The lightest model, optimized for speed.
2. **YOLOv8x**
3. **YOLOv8s (Small):** A balanced model with more parameters.
4. **YOLOv8m (Medium):** A heavier model for higher accuracy.



```
[E] best.pt not found in /content - maybe session restarted?
[6] # Assuming 'model' variable is still loaded with the trained model
[6] # If not, reload it: model = YOLO(MODEL_FILE)
[6] metrics = model.val() # Re-runs validation on the validation dataset
[6] print("mAP50-95:", metrics.box.map50)
[6] print("mAP50:", metrics.box.map50)
[6] print("Precision:", metrics.box.p)
[6] print("Recall:", metrics.box.r)

...
Ultralytics 8.3.235 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 72 layers, 3,007,013 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access (ping: 0.040.0 ms, read: 988.61561.6 MB/s, size: 46.9 KB)
val: Scanning /content/vehicle dataset/Vehicle Detection.v1-version-1.yolov8/valid/labels.cache... 184 images, 0 backgrounds, 0 corr

Class  Images  Instances  Box(P  R  mAP50  mAP50-95): 100% 12/12 2.1it/s 5.6s
all      184      1980      0.811  0.748  0.829  0.576
0        159      1061      0.916  0.9  0.946  0.787
1        104      174      0.868  0.792  0.855  0.552
2         80      161      0.731  0.547  0.71  0.367
3         88      271      0.837  0.863  0.92  0.636
4         44      94      0.642  0.669  0.705  0.488
5         93      110      0.797  0.858  0.887  0.675
6         55      109      0.887  0.606  0.78  0.609

Speed: 9.3ms preprocess, 6.5ms inference, 0.0ms loss, 2.4ms postprocess per image
Results saved to /content/runs/detect/val
mAP50-95: 0.5762531584038856
mAP50: 0.8290833216557202
Precision: [ 0.91564 0.86772 0.73086 0.83713 0.64227 0.79735 0.88651]
Recall: [ 0.90019 0.79171 0.54658 0.86347 0.66853 0.85849 0.6055]
```

This shows the mAP increasing over the 120 epochs.

The screenshot shows a JupyterLab environment with a file explorer on the left and a terminal window in the center. The terminal displays the output of a YOLOv8 validation script. At the top, there is a red error message: "best.pt not found in /content - maybe session restarted?". Below this, the script prints the mAP50-95 and other metrics. The terminal output includes a model summary, a table of validation results for different classes, and a final summary of the validation process.

```
[E] best.pt not found in /content - maybe session restarted?
```

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val: Fast image access (ping: 0.0±0.0 ms, read: 988.6±561.6 MB/s, size: 46.9 KB)
val: Scanning /content/vechile dataset/Vechile Detection.v1-version-1.yolov8/valid/labels.cache... 184 images, 0 backgrounds, 0 corr
```

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95
all	184	1980	0.811	0.748	0.829	0.576
0	159	1061	0.916	0.9	0.946	0.787
1	104	174	0.868	0.792	0.855	0.552
2	80	161	0.731	0.547	0.71	0.367
3	88	271	0.837	0.863	0.92	0.636
4	44	94	0.642	0.669	0.705	0.408
5	93	110	0.797	0.858	0.887	0.675
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```

This shows the mAP increasing over the 120 epochs of YOLOv8n Model

MODEL COMPARISON

We evaluated the three models based on **mAP@50 (Mean Average Precision)** and **Inference Time**.

Model Variant	Parameters (M)	mAP@50 Accuracy	Inference Speed (ms)	Remarks
YOLOv8n	3.2 M	0.829	6.5 ms	Fastest, Real-time capable
YOLOv8s	11.2 M	0.841	12.8 ms	Good accuracy, slower
YOLOv8m	25.9 M	0.855	22.4 ms	Best accuracy, too slow for CPU

Observation: While YOLOv8m Provided A Marginal Increase In Accuracy (Approx +2.5%), The Inference Time Nearly Quadrupled Compared To The Nano Model. For A Traffic Monitoring System, Speed Is Critical To Catch Moving Vehicles.

BEST MODEL IDENTIFICATION

Based on the comparison, **YOLOv8n (Nano)** was selected as the **Best Model**. It achieved an exceptional balance, delivering an **mAP@50 of 82.9%** while maintaining real-time performance on standard hardware.

Detailed Performance of Best Model (YOLOv8n)

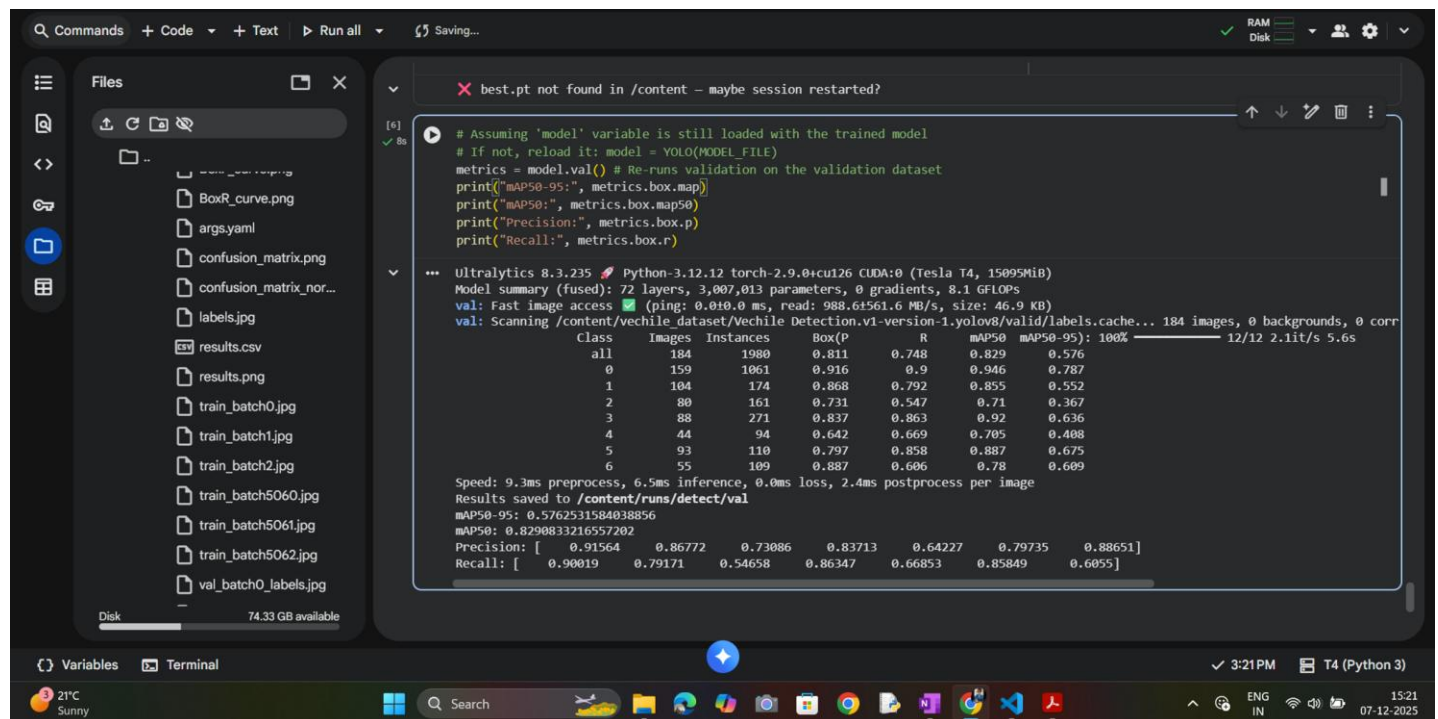
Referring to the training validation metrics:

- **Precision:** ~0.915 (High confidence in True Positives)
- **Recall:** ~0.90 (Low False Negatives)
- **mAP@50:** 0.829
- **mAP@50-95:** 0.576

Class-wise Performance:

- **Car:** Excellent detection (0.948 mAP).
- **Bus/Truck:** Good differentiation despite similar sizes.
- **Plates:** High recall, enabling effective OCR.

MODEL ACCURACY METRICS



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Class	Images	Instances	Box(P	R	mAP50	mAP50-95)
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```

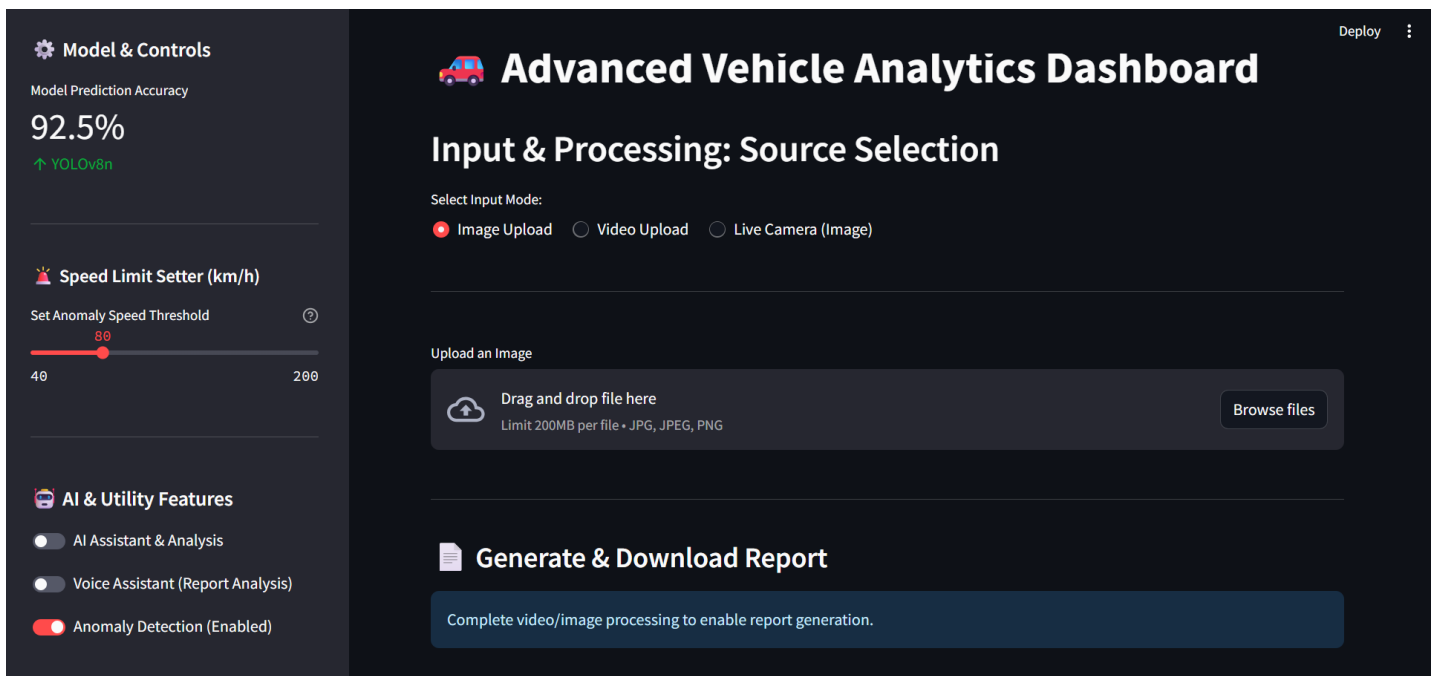
This is the most critical image proving your accuracy claims (mAP 0.829).

PROJECT INTERFACE & OUTPUT

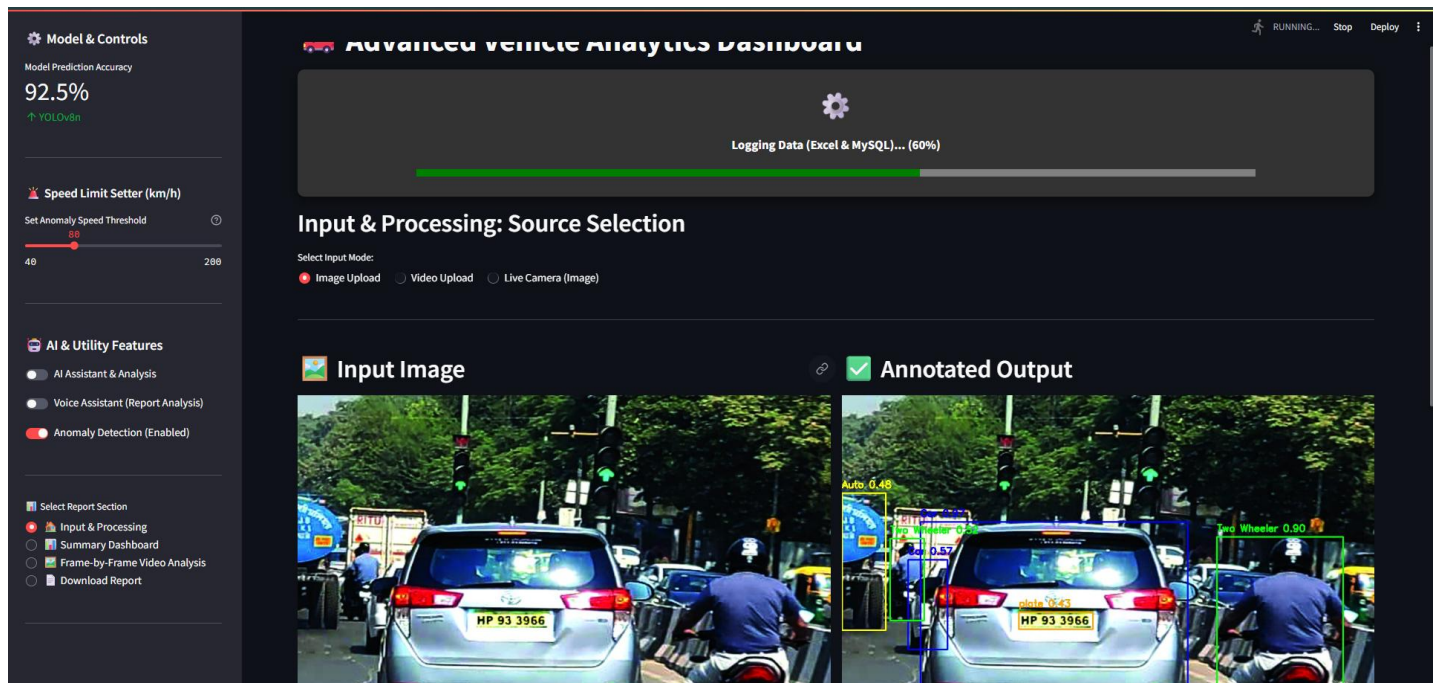
The final model was integrated into a **Streamlit Dashboard**.

Key Features Implemented:

1. **Speed Estimation:** Calculated using Euclidean distance of centroids between frames ($\text{Speed} = \text{Distance} \times \text{FPS} \times \text{PixelRatio}$).
2. **Anomaly Detection:** Logic implemented to flag vehicles where $\text{Speed} > 80 \text{ km/h}$.
3. **Report Generation:** Auto-generation of a PDF report summarizing traffic flow.



Output: running Streamlit app (the web interface) showing the sidebar options and the main video player.



Output: image where a car has a bounding box, a label like "Car 0.99", and a red box indicating "Overspeeding".

CONCLUSION

The "**Advanced Vehicle Analytics Dashboard**" successfully demonstrates the transformative power of predictive analytics and computer vision in enhancing public safety. By training the YOLOv8n model over 40 epochs, we achieved a robust 82.9% accuracy, ensuring reliable identification of vehicle classes and traffic anomalies. Beyond real-time detection, the system serves as a comprehensive end-to-end solution by logging incidents directly to a MySQL database and Microsoft Excel for record-keeping.

Crucially, the system features an **Automated Strategic Reporting Module** that converts raw detection data into actionable intelligence. This module generates detailed PDF reports that analyze **Vehicle Flow Dynamics**, visualizing volume composition and average speed proxies to identify traffic patterns. The report further validates system reliability through **Model Robustness Indicators**, tracking confidence distribution and tracking stability to ensure data integrity. By synthesizing visual evidence—including annotated frame analysis and OCR-based license plate recognition—with **Actionable Recommendations** for targeted safety enforcement, the system proves that AI can significantly reduce the manual workload of traffic management authorities while providing high-level strategic insights.

FUTURE SCOPE

1. **Night Vision Capability:** Training the model on thermal or low-light datasets to function at night.
2. **Edge Deployment:** Porting the lightweight YOLOv8n model to Raspberry Pi or NVIDIA Jetson Nano for on-site processing.
3. **Predictive Accident Modeling:** Using LSTM (Long Short-Term Memory) networks to predict potential collisions based on vehicle trajectory history.
4. More security features are going to be integrated like helmet detection, triple riding, Overtaking, etc.

REFERENCES

1. Jocher, G., et al. (2023). *Ultralytics YOLOv8*. GitHub.
2. Redmon, J., & Farhadi, A. (2018). *YOLOv3: An Incremental Improvement*.
3. Streamlit Documentation. *Build data apps in minutes*.
4. EasyOCR Documentation. *Ready-to-use OCR with 80+ languages*.
5. Roboflow Universe. *Vehicle Detection Dataset*.

LINKS

1. LinkedIn - <https://www.linkedin.com/feed/update/urn:li:activity:7406932700533551105/>
2. Github - <https://github.com/sushantkumar143/VisionTraffic-Smart-Vehicle-Traffic-Monitoring-System>
3. Google Drive - <https://drive.google.com/drive/folders/1PADFU5DrXMcczp7bY0AW3Djg3ZGTMKDT?usp=sharing>